

Low-Cost 3.7V To 5V-6V DC-TO-DC CONVERTER

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This article presents a simple, low-cost DC-to-DC converter that converts 3.6V-3.7V battery input into three outputs of 6V,

PARTS LIST

Semiconductors:

IC1	- MC34063A DC-to-DC converter control circuit
D1, D4	- 1N5819 Schottky diode
D2, D3	- 1N4007 rectifier diode
LED1, LED2	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 0.33-ohm/0.47-ohm
R2	- 180-ohm
R3	- 10-kilo-ohm
R4	- 2.7-kilo-ohm
R5, R6	- 3.3-kilo-ohm
R7	- 470-ohm

Capacitors:

C1 - 470 μ F, 16V electrolytic
C2, C4, C6, C8 - 100nF ceramic disk
C3 - 470pF ceramic disk
C5, C7, C9 - 100 μ F, 16V electrolytic

Miscellaneous:

- CON1-CON4 - 2-pin terminal connector
- F1 - 1A fuse with holder
- L1 - 100 μ H, 1A inductor
- 3.7V Li-ion battery or 3 $\times$ 1.2V NiCd batteries

5.3V and 5V. These voltage outputs can be used simultaneously or separately. The converter is useful for digital and analogue audio equipment including portable audio amplifiers and MP3 players, and for experiments with digital and analogue electronics.

3.7V and 3.6V (3x1.2V) rechargeable batteries are widely available at low cost. But most equipment operate off 5V or 6V. So you cannot use 3.7V and 3.6V batteries even if you use two 3.7V batteries (3.7x2 = 7.4V) or four 1.2V batteries (1.2x4 = 4.8V). Hence you need a small DC-to-DC converter to get 5V, 6V or both of these, depending on the intended application. The converter described here provides a simple solution using an MC34063A IC.

Advantages of MC34063

MC34063A is ideal for small projects like DC-to-DC converters as it offers the following advantages:

1. Low cost
2. Easy-to-use DIP8 package
3. Operating input voltage of 3V-40V DC from NiCd, NiMH or Li-ion rechargeable batteries
4. Low quiescent current
5. Current limiting feature for the input voltage
6. Output adjustable with the help of a simple resistor divider network or trimmer potentiometer
7. Typical operation frequency of around 45kHz, which can be increased to around 100kHz
8. Built-in reference voltage of 1.25V
9. Internal output transistor capable of switching up to 1.5A

Circuit and working

Fig. 1 shows circuit of the 3.7V to 5V-6V DC-to-DC converter. It is built around switching regulator MC34063A (IC1), two 1N5819 Schottky diodes (D1 and D4), two 1N4007 rectifier diodes (D2 and

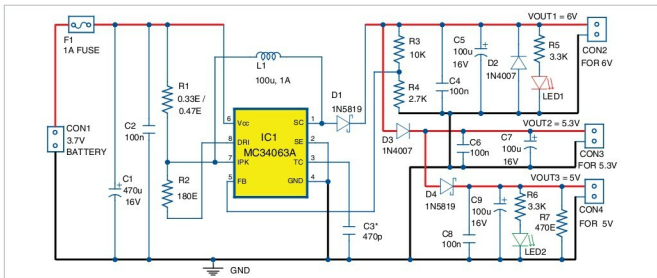


Fig. 1: Circuit diagram of low-cost 3.7V to 5V-6V DC-to-DC converter